



ANDERSON JUNIOR COLLEGE
2017 JC 2 PRELIMINARY EXAMINATIONS

NAME: _____

PDG: _____/16

CHEMISTRY

Paper 2 Structured Questions

9729/02

13 September 2017

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use		
Paper 2	1	/ 15
	2	/ 13
	3	/ 16
	4	/ 14
	5	/ 17
Total		/ 75

This document consists of **20** printed pages.

Answer **all** the questions.

- 1 (a) *Use of the Data Booklet will be relevant to this question.*

Iron ore from different mines will contain different percentages by mass of iron. The percentage of iron in a sample of ore can be estimated by converting all of the iron present into $\text{Fe}^{2+}(\text{aq})$ ions and then using a redox titration.

The sample of ore is crushed, weighed and then dissolved in aqueous acid. The $\text{Fe}^{3+}(\text{aq})$ ions are then reduced to $\text{Fe}^{2+}(\text{aq})$ ions by reaction with an excess of aqueous tin(II) chloride, SnCl_2 .

- (i) Construct an ionic equation for the reduction of $\text{Fe}^{3+}(\text{aq})$ ions with aqueous tin(II) chloride solution.

.....[1]

- (ii) Calculate the standard cell potential, $E_{\text{cell}}^{\ominus}$, for this reaction.

[1]

- (iii) Explain why tin(II) chloride will reduce $\text{Fe}^{3+}(\text{aq})$ ions to $\text{Fe}^{2+}(\text{aq})$ ions, but will **not** reduce $\text{Fe}^{2+}(\text{aq})$ ions to metallic iron.

.....

[2]

A sample of iron ore weighing 11.05 g was converted to $\text{Fe}^{2+}(\text{aq})$ ions using the method described above. The resultant solution was then made up to a volume of 250 cm^3 in a volumetric flask.

25.0 cm^3 portions of this solution were then titrated with $0.100 \text{ mol dm}^{-3}$ of aqueous potassium dichromate(VI) using a suitable indicator. The results are shown below.

titration number	1	2	3
initial burette reading / cm^3	0.00	19.95	2.10
final burette reading / cm^3	19.95	40.05	22.15
titre / cm^3	19.95		

- (iv) Complete the table on page 2 and use the results to determine the number of moles of potassium dichromate(VI) required to react with the Fe^{2+} ions in 25.0 cm^3 of the solution.

[2]

- (v) Write an ionic equation for the reaction of Fe^{2+} ions with acidified $\text{Cr}_2\text{O}_7^{2-}$ ions.

.....[1]

- (vi) Calculate the total number of moles of Fe^{2+} in the original solution made up from the iron ore, and hence calculate the percentage by mass of iron in the sample of iron ore.

[2]

(b) Iron and magnesium form iodate(V) compounds as shown, $\text{Fe}(\text{IO}_3)_2$ and $\text{Mg}(\text{IO}_3)_2$. Upon heating, each of the iodates decomposes to give a solid residue. Purple fumes and a gas that relights a glowing splint are also observed.

- (i)** Draw a dot-and-cross diagram to show the bonding in iodate(V) ion, IO_3^- . In your diagram use the symbols '•' and 'x' to distinguish electrons from iodine and oxygen atoms, and the symbol '□' for any additional electrons responsible for the overall negative charge.

[1]

- (ii)** Write a balanced equation for the thermal decomposition of $\text{Mg}(\text{IO}_3)_2$.

.....[1]

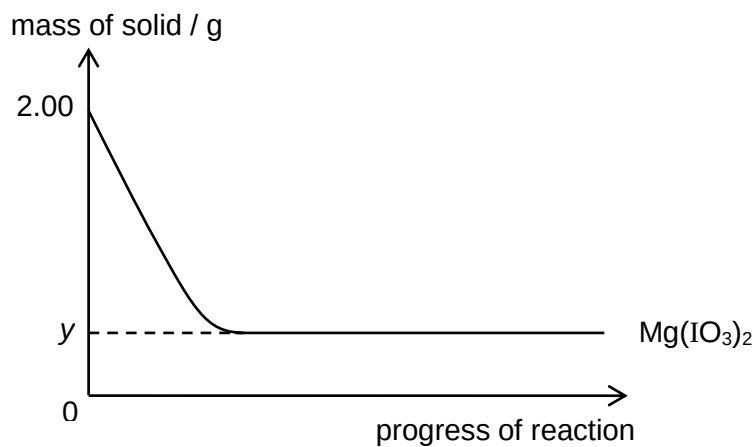
- (iii)** $\text{Mg}(\text{IO}_3)_2$ is found to decompose at 900 K.

By quoting suitable data from the *Data Booklet*, suggest and explain if $\text{Fe}(\text{IO}_3)_2$ will decompose at a higher or lower temperature than $\text{Mg}(\text{IO}_3)_2$.

.....

[2]

The graph below shows the change in mass when 2.00 g of $\text{Mg}(\text{IO}_3)_2$ is heated at temperature, $T^\circ\text{C}$.



(iv) Determine the value of y .

[1]

(v) On the same axes, show clearly the sketch for the decomposition of 2.00 g of $\text{Fe}(\text{IO}_3)_2$ at the same temperature.

[1]

[Total: 15]

- 2 (a) Explain the meaning of the term *homogenous catalyst* and illustrate your answer by using one specific reaction involving a transition metal or its compound behaving as one.

.....
.....
.....
.....[2]

- (b) *Use of the Data Booklet is relevant to this question.*

Aqueous hydrogen peroxide is fairly stable but when a mixture of cobalt(II) salt and tartaric acid $\text{HO}_2\text{CCH}(\text{OH})\text{CH}(\text{OH})\text{CO}_2\text{H}$ is added to aqueous hydrogen peroxide, the following changes take place.

The initial pink solution *slowly* turns into a green solution containing cobalt(III) species and then oxygen is *vigorously* evolved. Finally the solution turns pink again. Tartaric acid and the cobalt(II) salt can be recovered from the solution at the end.

- (i) Write an equation for the overall reaction.

.....[1]

- (ii) State the role of cobalt(II) salt in this reaction and give evidence to support your answer.

.....
.....
.....[2]

- (iii) With the aid of relevant data in the *Data Booklet*, show that $\text{Co}^{3+}(\text{aq})$ is not stable in aqueous solution and hence comment on the role of tartaric acid in this reaction.

.....
.....
.....
.....
.....[2]

- (c) The decomposition of hydrogen peroxide follows a first order kinetics.

A 100 cm³ of hydrogen peroxide undergoes complete decomposition to liberate 96 cm³ of oxygen gas. It is found that 48 cm³ of oxygen gas was collected in 35 minutes. All gaseous volumes were measured at room temperature and pressure.

- (i) Calculate the initial concentration of hydrogen peroxide.

[1]

- (ii) Estimate the time taken for 88 cm³ of oxygen gas to be collected.

[1]

- (iii) Calculate the rate constant k , including the units for the reaction.

[2]

- (iv) What percentage of the hydrogen peroxide in a solution of concentration 0.20 mol dm⁻³ would decompose in 35 minutes at the same temperature?

[1]

- (v) Outline a suitable method, other than the gas collection method, to find the order of reaction for the decomposition of hydrogen peroxide.

.....

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.....[1]

[Total: 13]

- 3 The Pollutant Standards Index (PSI) is an air quality indicator. It is based on five pollutants: particulate matter (PM10), sulfur dioxide, carbon monoxide, ozone and nitrogen dioxide.

To calculate the overall PSI, the PSI value is first determined, using the following table, for **each** of the five pollutants.

<i>i</i>	PSI value, P_i	Concentration, C_i				
		PM10 ($\mu\text{g m}^{-3}$)	SO ₂ ($\mu\text{g m}^{-3}$)	CO (mg m^{-3})	Ozone ($\mu\text{g m}^{-3}$)	NO ₂ ($\mu\text{g m}^{-3}$)
1	50	50	80	5.0	118	–
2	100	150	365	10.0	157	–
3	200	350	800	17.0	235	1130
4	300	420	1600	34.0	785	2260
5	400	500	2100	46.0	980	3000
6	500	600	2620	57.5	1180	3750

[1 μg = 10^{-6} g; 1 mg = 10^{-3} g]

Given the concentration of a pollutant (with units as stated in the table above), where C_{i+1} > concentration of pollutant > C_i ,

$$\text{PSI of pollutant} = \left[\left(\frac{P_{i+1} - P_i}{C_{i+1} - C_i} \right) (\text{concentration of pollutant} - C_i) \right] + P_i$$

The overall PSI is then based on the maximum value out of the five calculated PSI pollutant values.

In Singapore, the 24-hr PSI is used by the National Environment Agency (NEA) to provide health advisory.

24-hr PSI	Healthy Persons	Elderly, Pregnant women, Children	Persons with chronic lung disease, heart disease, stroke
< 100	Normal activities		
101–200	Minimise prolonged or strenuous outdoor physical exertion.	Minimise prolonged outdoor activity.	Avoid all outdoor activities. If outdoor activity is unavoidable, wear N95 mask.
201–300	Avoid prolonged or strenuous outdoor physical exertion. If outdoor activity is unavoidable, wear N95 mask.	Avoid all outdoor activities.	
>300	Minimise all outdoor exposure. If outdoor activity is unavoidable, wear N95 mask.	If outdoor activity is unavoidable, wear N95 mask (for adults).	

- (a) (i) In a 1 m^3 sample of air, the mass of PM₁₀ and carbon monoxide were found to be $320 \text{ } \mu\text{g}$ and 20 mg respectively. Calculate the PSI values for each of the above pollutants, leaving your answers to **3 significant figures**.

[2]

- (ii) Given that the PSI value of sulfur dioxide, ozone and nitrogen dioxide are 150, 112 and 133 respectively for the same sample of gas, use these values and your answers to (a)(i) to determine the overall PSI.

[1]

- (iii) Assuming that the current overall PSI level is your answer in (a)(ii), what advice would you give to a Physical Education (PE) teacher in Anderson Junior College who will be conducting a PE lesson soon?

.....

.....[1]

- (b) The amount of sulfur dioxide in a sample of air can be determined by first reacting it with sodium iodate, NaIO_3 . Iodine is one of the products in this reaction.

(i) Write an ionic equation for the reaction between sulfur dioxide and sodium iodate.

.....[1]

- (ii) When a 1 m^3 sample of air was bubbled through a solution of sodium iodate, the resulting solution was neutralised by 10.0 cm^3 of $0.005 \text{ mol dm}^{-3}$ sodium hydroxide solution.

Calculate the concentration of sulfur dioxide, in $\mu\text{g m}^{-3}$, in the sample of air.

[2]

- (c) Some countries have set limits for particulates in the air. For example, the European Union has a daily average (24-hour) limit of $180 \mu\text{g m}^{-3}$ for PM10. Cities that violate this daily limit face a hefty financial penalty that is calculated with consideration of many factors such as the severity the violation has on the ecosystem, duration of the non-compliance and country's GDP etc.
- (i) A collected sample of air from the German city of Leipzig contains $2 \times 10^{-5} \%$ by mass of PM10. Given that the density of air is 1 kg m^{-3} , calculate the concentration of PM10 in the sample collected.

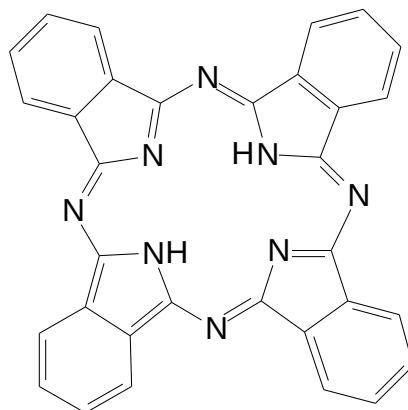
[1]

- (ii) Hence, deduce whether the German city of Leipzig will be faced with any financial penalty.

.....

.....[1]

- (d) A team of researchers at the National University of Singapore has come up with an eco-friendly and cost-effective solution to keep out the harmful PM_{2.5} particles found in haze. They created a nanofibre liquid solution by modifying phthalocyanine, a blue-green coloured aromatic compound commonly used in dyes for clothes. The solution can be applied to non-woven mesh to become thin, see-through air filters that block out up to 90% of PM_{2.5} particles.



phthalocyanine

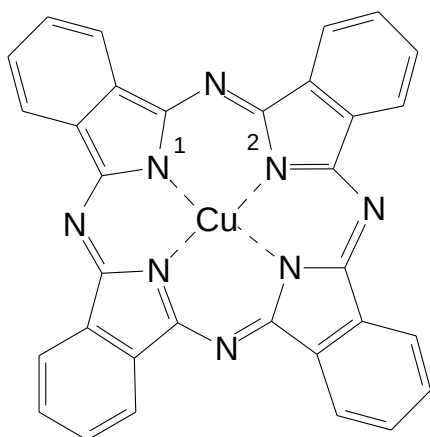
- (i) Give the molecular formula of phthalocyanine.

.....[1]

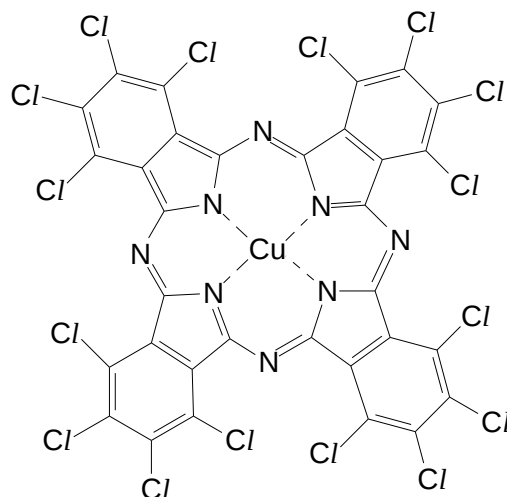
- (ii) Suggest why phthalocyanine can be described as aromatic.

.....[1]

- (e) Copper phthalocyanine dyes are artificial organic pigments derived from phthalocyanine. They are produced by introducing solubilising groups, such as methanesulfonic acid, $\text{CH}_3\text{SO}_3\text{H}$.



phthalocyanine blue



phthalocyanine green

- (i) Name the type of bonds formed between copper and the nitrogen atoms numbered 1 and 2 in phthalocyanine blue.

N1

N2

[2]

- (ii) Give the electronic configuration of the copper in phthalocyanine blue.

.....[1]

- (iii) Deduce the number of electrons around the N1 nitrogen atom.

.....[1]

- (iv) Account for the difference in the colour of the two copper phthalocyanine dyes given above.

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.....[1]

[Total: 16]

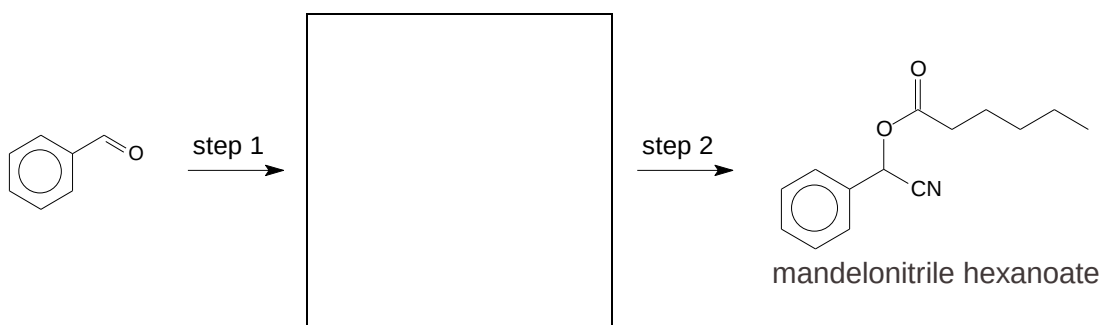
- 4 The *Oribatula tibialis* mite uses hydrogen cyanide, HCN, a highly volatile and toxic substance, to poison its predators.

The HCN is stored in the form of mandelonitrile hexanoate, $C_{14}H_{17}NO_2$, in the mite's oil glands, to avoid poisoning itself.

- (a) In order to confirm the identity of mandelonitrile hexanoate in mite's oil glands, synthetic mandelonitrile hexanoate was made in the laboratory and dissolved in hexane.

Both synthetic mandelonitrile hexanoate and the extracted oil gland from *Oribatula tibialis* mite were then sent for analysis using various analytical techniques.

- (i) Suggest the reagents that can be used to prepare mandelonitrile hexanoate from benzaldehyde at **low** temperature conditions. Show the structure of the intermediate organic compound in the box below.



step 1

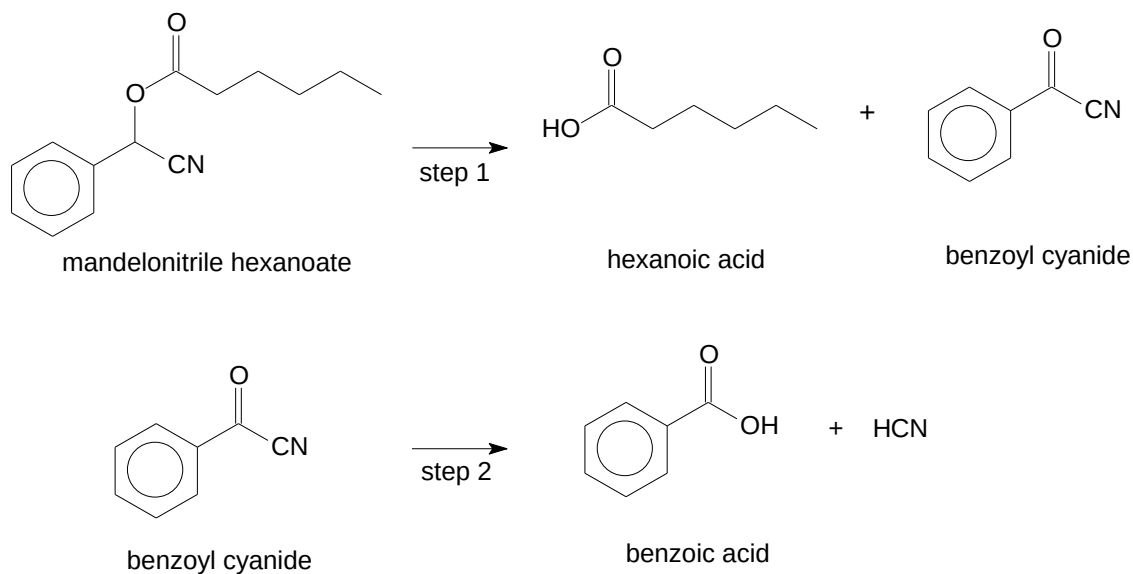
step 2 [3]

- (ii) Suggest, in terms of the intermolecular forces involved, a reason why mandelonitrile hexanoate can dissolve in hexane.

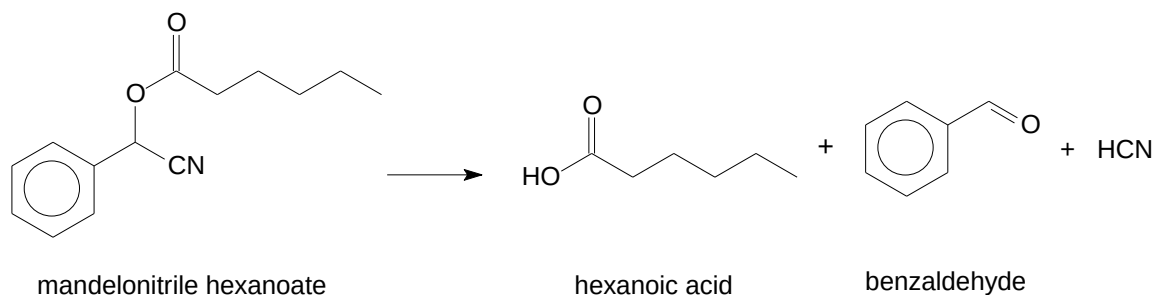
..... [1]

When attacked by predators, the *Oribatula tibialis* mite secretes mandelonitrile hexanoate, which then release HCN, when in contact with the moisture, for example, from the predators' saliva. The HCN can be released via two different chemical pathways.

Pathway I illustrates a two-step process.



In pathway II, benzaldehyde is produced instead of benzoic acid.



- (b) (i) By considering the change in structural formula shown in pathway I, suggest **two** types of reaction that could have occurred in step 1.

.....
[2]

- (ii) In step 2 of pathway I, the intermediate benzoyl cyanide breaks down into benzoic acid and HCN.

It was reported that benzoyl cyanide is less reactive than benzoyl chloride when it is in contact with water. Suggest a possible reason for this.

.....
[1]

- (iii) Suggest a simple chemical test that could be carried out to confirm which chemical pathway has occurred.

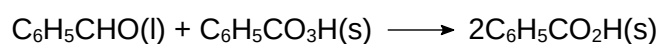
You may assume that HCN has vapourised completely from the product mixture.

.....

[2]

- (c) Benzaldehyde can be oxidised by peroxybenzoic acid, without consumption of oxygen, to give benzoic acid.

The standard enthalpy change of the reaction below is -316 kJ mol^{-1} .



The standard enthalpy change of formation of peroxybenzoic acid and benzaldehyde are given in the table below.

Name	Formula	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
peroxybenzoic acid	$\text{C}_6\text{H}_5\text{CO}_3\text{H(s)}$	-367
benzaldehyde	$\text{C}_6\text{H}_5\text{CHO(l)}$	-87

- (i) Define the term *standard enthalpy change of formation*, $\Delta H_f^\ominus$ of a compound.

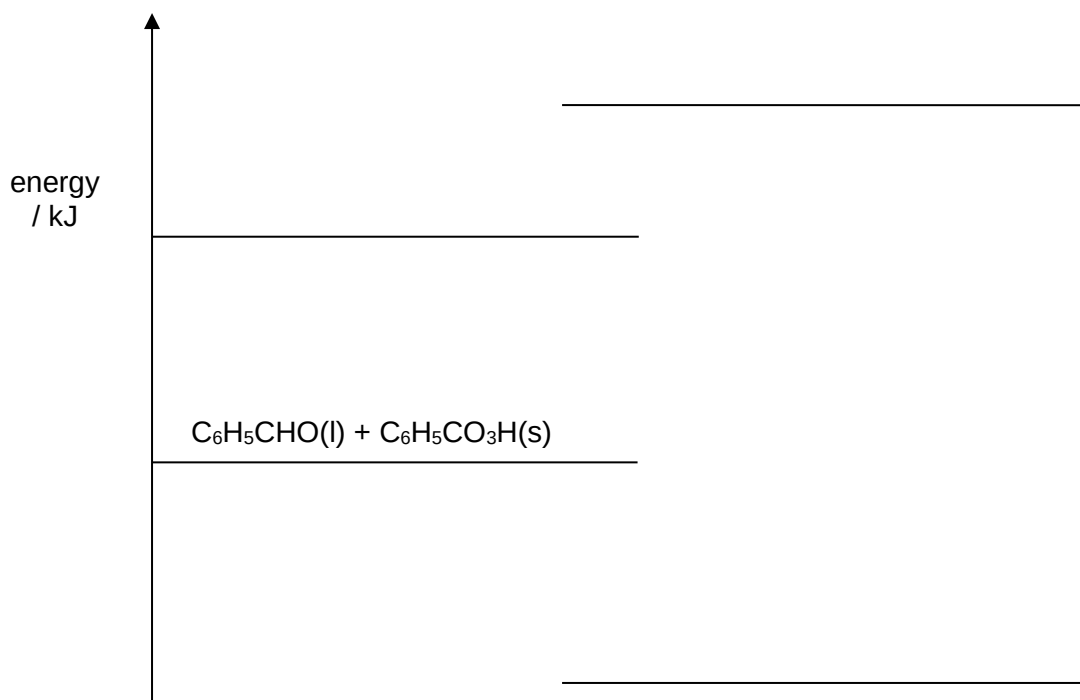
.....
[1]

- (ii) Use the data given above to calculate the enthalpy change of formation of benzoic acid.

[1]

- (iii) Draw arrows on the energy diagram below to show each of the energy terms involved in the steps you have used in your calculation in (c)(ii).

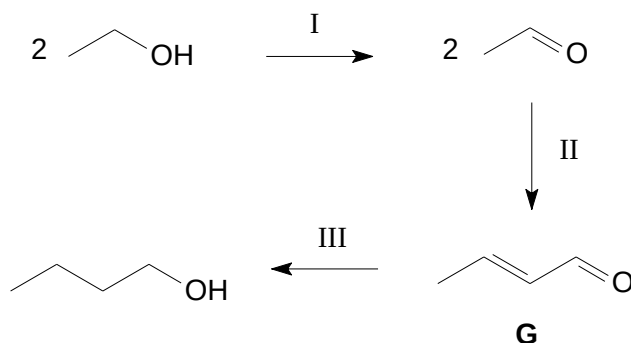
Label each level with the appropriate formulae.



[3]

[Total: 14]

- 5 Scientists in the United States of America have come up with a simple 3-stage process to convert ethanol into butan-1-ol, in what could be an important step forward for renewable energy.



- (a) (i) State the reagents and conditions required for stages I and III.

stage I

stage III

[2]

- (ii) Write an equation for the incomplete combustion of butan-1-ol, giving CO_2 and CO in a 3 : 1 ratio.

.....[1]

- (iii) Describe the type(s) of stereoisomerism shown by compound **G**.
Draw the displayed formula of the stereoisomers of **G**.

Type of isomerism

[2]

- (iv) **G** reacts with a suitable reducing agent to give compound **H**.

H effervesces with sodium metal but does not react with 2,4-dinitrophenylhydrazine.
H also decolourises aqueous bromine.

Suggest the structure of **H** and explain these observations.

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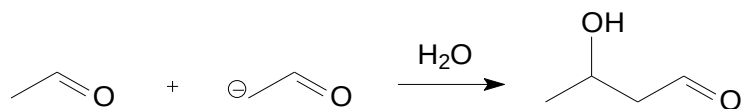
.....[2]

- (b) The mechanism for stage II of the process in forming the α,β -unsaturated aldehyde **G** is thought to involve three steps.

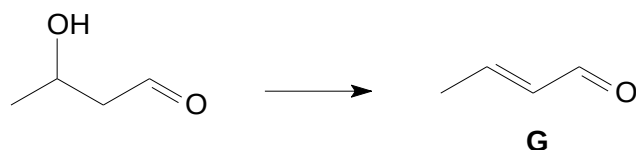
Step 1 The first step involves an acid–base reaction.



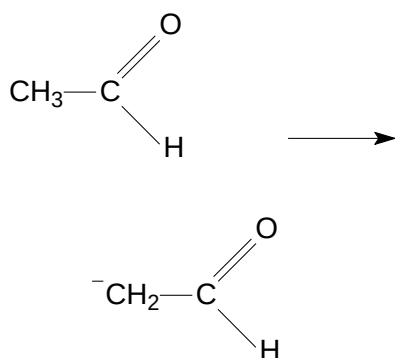
Step 2 This is followed by a nucleophilic addition reaction to form the intermediate 3-hydroxybutanal.



Step 3 Elimination of water is involved in the last step.



- (i) Complete the diagram to suggest a mechanism to show how 3-hydroxybutanal intermediate is formed. Show all charges and relevant lone pairs and show the movement of electron pairs by using curly arrows.



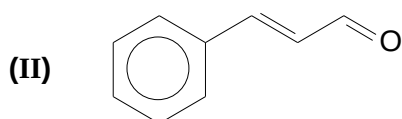
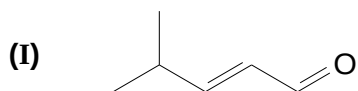
[2]

- (ii) A different product can be formed together with **G**, in step 3, albeit in lower yield. Suggest a reason for this.

.....

[1]

- (iii) Deduce the structures of suitable carbonyl compounds to synthesise each of the following α,β -unsaturated aldehyde.



[3]

- (c) Butan-1-ol has a number of constitutional isomers.

- (i) Draw the structural formula of the other three alcohols with the same molecular formula as butan-1-ol. Label your structures **J**, **K** and **L**. Classify these alcohols as primary, secondary or tertiary.

[2]

- (ii) Identify which alcohol reacts with alkaline aqueous iodine and write a balanced equation for the reaction, showing the structural formula of the products.

[2]

[Total: 17]